

MUJTABA TARIQ

AI / Full-Stack Developer | LLM Fine-Tuning | Automation Systems

Islamabad, Pakistan | (+92) 316 5423590 | mujtabatariq01@gmail.com

linkedin.com/in/mujtabatariq09 | github.com/MujtabaIT2002 | mujtabatariq.dev

Summary

AI and Full-Stack Developer with 2+ years building production-grade machine learning, LLM, and automation systems across healthcare, e-commerce, and bioinformatics domains. Proven ability to ship end-to-end solutions, from fine-tuning Phi-3 Mini on consumer-grade hardware via QLoRA to architecting multi-source data pipelines integrating Shopify, third-party APIs, and local LLM inference. BS in Information Technology from Bahria University. Seeking applied ML, AI Engineering, or LLM tooling roles where I can deliver measurable business impact.

Skills

Core: Python, JavaScript, React.js, Node.js, PyTorch, Transformers, PEFT, QLoRA, REST APIs, PostgreSQL, Docker

AI / Machine Learning: TensorFlow, Scikit-learn, OpenCV, XGBoost, KNN, K-means, CNNs, LLM Fine-Tuning, 4-bit Quantization, Model Deployment

Web & Backend: Express.js, Flask, Streamlit, JWT Authentication, Prisma ORM, MongoDB, MySQL

Automation & Integration: Playwright, Web Scraping, Google Apps Script, Shopify Admin API, Meta Graph API, Google Maps API

Tools & Platforms: Git, GitHub, Docker, Postman, Cloudinary, CUDA, Hugging Face Hub

Languages: English (Professional), Urdu (Native)

Experience

Data Annotator Jan 2026 – Present
Motive Islamabad, Pakistan

- Designed an internal workflow optimization process that improved task allocation efficiency and lifted overall pod throughput by an estimated [70%], adopted across the annotation team.
- Contribute to training data pipelines for fleet computer vision models by annotating large-scale video datasets with bounding boxes for object detection, scene understanding, and motion tracking.
- Drive frame-level quality assurance across dynamic visual scenes, improving annotation consistency and reducing downstream model retraining cycles.
- Operate within a pod-based delivery model, consistently meeting SLA targets for throughput, accuracy, and labeling guideline compliance.

AI Solutions Developer Jun 2025 – Sep 2025
Rebiz (Oxiliry) Remote / Pakistan

- Architected and shipped AI-driven automation systems covering daily summaries, email triage, uptime monitoring, and weekly executive reporting, reducing manual reporting effort by an estimated [5 hours/week].
- Engineered a multi-source data pipeline unifying Shopify Admin API, search feeds, news APIs, and local LLM inference to produce structured operational intelligence reports.
- Delivered a competitor price-matching and normalization engine that powered pricing intelligence dashboards, tracking SKUs across competitor sources.
- Containerized backend workflows with Docker and exposed them via Flask REST APIs, enabling reliable handoff between automation services and downstream dashboards.

MERN Stack Developer Mar 2025 – May 2025
Macro Soar Pakistan

- Shipped core features of ormo.ai, an AI-powered consultancy platform enabling consultants to register and deploy custom AI chatbots without writing code.
- Built responsive React.js interfaces for consultant onboarding, chatbot configuration, and end-user workflows, increasing self-service completion rates and reducing support load.
- Developed Node.js and MongoDB backend services managing users, consultancy sessions, and chatbot metadata, with a focus on scalability and data integrity.

Front-End Developer Intern Sep 2024 – Feb 2025
Digital Empowerment Pakistan

- Built a production-ready React-based search engine integrating an external API, structured around reusable components and predictable state management.
- Leveraged React Hooks and the Context API to centralize global state, optimize data fetching, and cut redundant re-renders for a smoother user experience.
- Implemented dynamic form handling and client-side validation flows, improving input reliability and reducing invalid submissions.

- Led a small engineering team in delivering Google Workspace automation solutions across Drive, Sheets, Gmail, and Calendar, eliminating recurring manual workflows.
- Engineered Google Apps Script automations integrated with third-party APIs, extending internal tooling and consolidating disparate workflows into a single automation layer.
- Built dynamic web forms wired into Google Sheets for real-time data capture, tracking, and reporting, improving data consistency and reducing administrative effort.

Projects

Phi-3 Mini Marketing AI Assistant

LLM Fine-Tuning / QLoRA

- Fine-tuned Phi-3 Mini 4k-Instruct into a domain-specific marketing assistant (captions, ad copy, slogans, brand content) on a curated dataset of 400 structured samples, achieving noticeably improved tone control and brand-specific phrasing over the base model.
- Engineered the full training stack on a consumer 4GB GTX 1650 using 4-bit NF4 quantization, QLoRA, PEFT, gradient checkpointing, and paged AdamW, demonstrating production-grade fine-tuning under tight VRAM constraints.
- Stack: Python, Hugging Face Transformers, PEFT, QLoRA, Phi-3 Mini, 4-bit Quantization, CUDA.

Smart Construction Model

Full-Stack ML System

[GitHub Repository](#)

- Designed and shipped a full-stack platform for construction cost estimation, real estate recommendation, and society analytics, unifying ML inference and a web frontend behind a single user experience.
- Built an XGBoost-based price prediction engine alongside a K-means + KNN recommendation system leveraging Google Maps geolocation data, surfacing relevant societies and price trends from historical records.
- Stack: React.js, Node.js, Flask, PostgreSQL, Prisma ORM, XGBoost, K-means, KNN, Google Maps API, JWT.

Medical AI Assistant Bot

AI Assistant / Healthcare Automation

- Built an AI-powered medical triage assistant that collects symptoms, asks severity-based follow-ups, and routes users to general guidance, urgent care messaging, or doctor recommendations.
- Implemented a Playwright-based scraping workflow to ingest top doctor data from Marham, powering a doctor discovery and matching layer integrated with the conversational flow.
- Stack: Python, Web Scraping, LLM Integration, REST APIs.

Protein Antigenicity Prediction System

Bioinformatics / Deep Learning

[GitHub Repository](#)

- Developed a Flask-based web application predicting protein antigenicity from FASTA sequence input, combining UniRep-style protein embeddings with a CNN classifier for immunogenicity prediction.
- Containerized the application with Docker and Gunicorn into a deployment-ready service, exposing prediction confidence and results through a responsive web interface.
- Stack: Python, Flask, PyTorch, TensorFlow, CNN, UniRep Embeddings, FASTA, Docker.

X-ray Image Classification

Computer Vision / Machine Learning

- Built an end-to-end web application classifying chest X-ray images as Normal or COVID-19, integrating an OpenCV preprocessing pipeline (grayscale conversion, resizing, normalization) with a trained KNN classifier.
- Surfaced model evaluation outputs (accuracy, confusion matrix, ROC curve) directly in the UI, enabling transparent model inspection alongside real-time predictions.
- Stack: Python, Flask, OpenCV, Scikit-learn, KNN.

Additional Projects

Automation / API Integration / Scraping

- [Solace Pixel](#): Streamlit tool that generates portrait-format social posts, uploads to Cloundinary, and publishes directly to Instagram and Facebook via the Meta Graph API.
- [Terminal-Based Daraz Scraper](#): Playwright-driven CLI that extracts product titles, prices, ratings, reviews, screenshots, and structured JSON for market and price intelligence.
- Prayer Time Calculation Engine: CLI engine computing Fajr, Dhuhur, Asr, Maghrib, and Isha timings from astronomical inputs (solar declination, equation of time, hour angles) with zero external dependencies.

Education

Bahria University

Mar 2021 – Feb 2025

Bachelor of Science in Information Technology | CGPA: 3.25 / 4.00

Islamabad, Pakistan

- Relevant coursework: Data Structures, Object-Oriented Programming, Database Design and Management, Software Requirements Engineering.